Faculty: FTMI
Course: Введение в веб технологии
Year: 2025/2026
Group: U4225
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Lab: Lab1
Date of create: 13.03.2026
Date of finished: 15.03.2026

Лабораторная работа №3

Ход работы

1. Создал папку Prometheus с yml файлом в гите и на маке, но была трудность при доступах терминала к документам. По итогу получилось с помощью Сергея!

main devops-lab-malikov / prometheus / prometheus.yml

artemmalickov2015-cmyk Rename lab3/prometheus/prometheus.yml to prometheus/p

Code Blame

```yaml
global:
  scrape_interval: 15s

scrape_configs:
  - job_name: 'prometheus'
    static_configs:
      - targets: ['localhost:9090']

  - job_name: 'node-exporter'
    static_configs:
      - targets: ['node-exporter:9100']
```

2. После фикса ошибки запустил контейнер Node-exporter и проверена работа на хосте

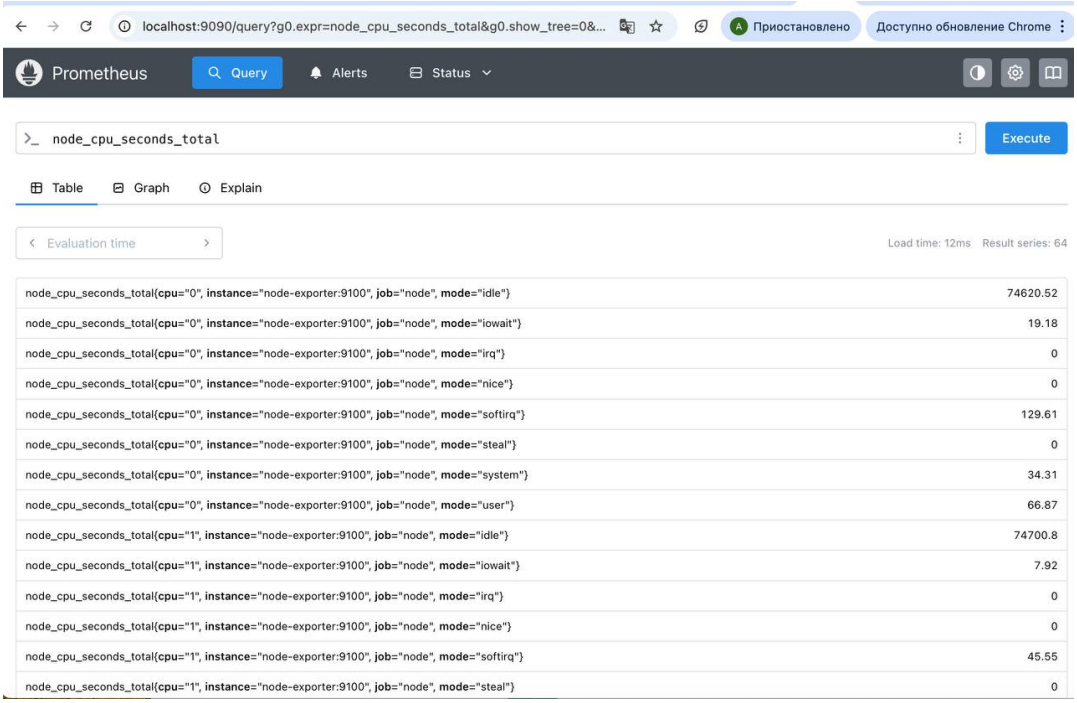
```
flocktory@MacBook-Air-Flocktory-69 ~ % docker run -d \
      --name node-exporter \
      --restart=unless-stopped \
      -p 9100:9100 \
      -v "/proc:/host/proc:ro" \
      -v "/sys:/host/sys:ro" \
      -v "/:/rootfs:ro" \
      prom/node-exporter \
      --path.procfs=/host/proc \
      --path.rootfs=/rootfs \
      --path.sysfs=/host/sys \
      --collector.filesystem.mount-points-exclude="^/(sys|proc|dev|host|etc)($$|/)"

4c238f80a7b92ffab50223fa02246a50685dfdd0a09b940c2f44a8189775130d
flocktory@MacBook-Air-Flocktory-69 ~ % curl http://localhost:9100/metrics
# HELP go_gc_duration_seconds A summary of the wall-time pause (stop-the-world) duration in garbage collection cycles.
# TYPE go_gc_duration_seconds summary
go_gc_duration_seconds{quantile="0"} 0
go_gc_duration_seconds{quantile="0.25"} 0
go_gc_duration_seconds{quantile="0.5"} 0
go_gc_duration_seconds{quantile="0.75"} 0
go_gc_duration_seconds{quantile="1"} 0
go_gc_duration_seconds_sum 0
go_gc_duration_seconds_count 0
```

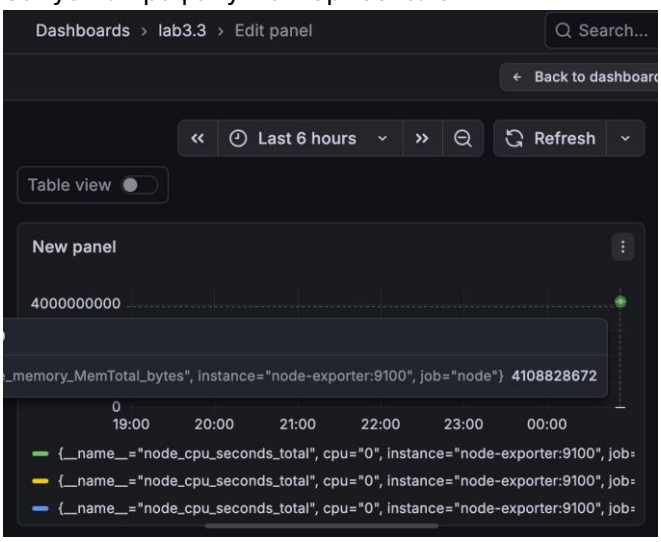
3. Создал том данных и сеть мониторинга

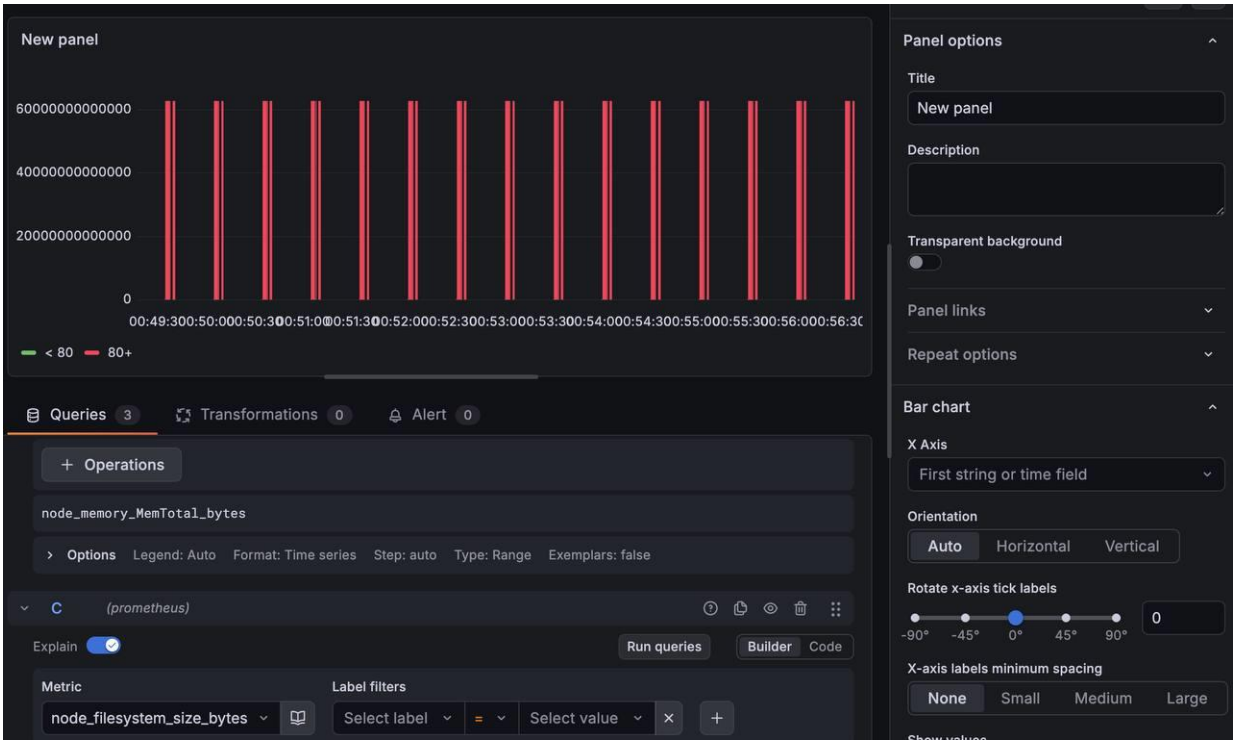
```
b6932ac909224ec36911c77c60b7971a3ec7c8a5209356e1a47e16b4315fa6ba
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker ps -a
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS                     PORTS                                          NAMES
b6932ac90922   prom/prometheus   "/bin/prometheus --c…"   15 seconds ago Restarting (2) 1 second ago                                            prometheus
5711cd18bba5   prom/node-exporter "/bin/node_exporter …"  45 minutes ago Up 45 minutes              0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp    node-exporter
44b1f21312a4   ubuntu            "bash"                   3 days ago     Up 3 days                                                                 volume-test2
43750eaa24ae   ubuntu            "bash"                   3 days ago     Up 3 days                                                                 volume
b8dc6d541e31   ubuntu            "bash"                   4 days ago     Exited (0) 4 days ago                                                     exciting_lichterman
2c08209d64b3   hello-world       "/hello"                 4 days ago     Exited (0) 4 days ago                                                     eager_swirles
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % nano prometheus/prometheus.yml
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % cd ~/monitoring-lab
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker ps -a
CONTAINER ID   IMAGE             COMMAND                  CREATED        STATUS                     PORTS                                          NAMES
b6932ac90922   prom/prometheus   "/bin/prometheus --c…"   3 minutes ago  Up 6 seconds               0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp    prometheus
5711cd18bba5   prom/node-exporter "/bin/node_exporter …"  49 minutes ago Up 48 minutes              0.0.0.0:9100->9100/tcp, [::]:9100->9100/tcp    node-exporter
44b1f21312a4   ubuntu            "bash"                   3 days ago     Up 3 days                                                                 volume-test2
43750eaa24ae   ubuntu            "bash"                   3 days ago     Up 3 days                                                                 volume
b8dc6d541e31   ubuntu            "bash"                   4 days ago     Exited (0) 4 days ago                                                     exciting_lichterman
2c08209d64b3   hello-world       "/hello"                 4 days ago     Exited (0) 4 days ago                                                     eager_swirles
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker volume create grafana-data
grafana-data
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker run -d \
      --name grafana \
      --network monitoring \
      --restart=unless-stopped \
      -p 3000:3000 \
      -v grafana-data:/var/lib/grafana \
      -e "GF_SECURITY_ADMIN_PASSWORD=admin" \
      grafana/grafana
Unable to find image 'grafana/grafana:latest' locally
latest: Pulling from grafana/grafana
ab4a0927d543: Pull complete
1dee255ae328: Pull complete
5a20d12d9f76: Pull complete
0091530ecb3a: Pull complete
bed6b68f7b55: Pull complete
90308bbf799b: Pull complete
7e9d2b4848b2: Pull complete
d8ad8cd72600: Pull complete
4f4fb700ef54: Pull complete
b63651784c9a: Pull complete
5ece6d253f06: Pull complete
Digest: sha256:e932bd6ed0e026595b08483cd0141e5103e1ab7ff8604839ff899b8dc54cabcb
Status: Downloaded newer image for grafana/grafana:latest
06919a8fdad59e05085fb261aa0e2aeac24b2c2b35cac409e2784d3c1e15f894
```

4. Проверка работы – запустился прометеус и подтянул в него данные



5. Запустил Графану и авторизовался



6. Сначала данные не собирались, провел мониторинг через команду inspect – данные появились, и добавил три метрики для визуализации. Сохранил дашборд

```
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker network connect monitoring node-exporter
flocktory@MacBook-Air-Flocktory-69 monitoring-lab % docker network inspect monitoring
[
    {
        "Name": "monitoring",
        "Id": "23ed428375bb24186d0c55f54de7746ccad5e1759fa5f43f47ec66aaabd83c84",
        "Created": "2026-03-16T14:28:48.981273626Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": {},
            "Config": [
                {
                    "Subnet": "172.18.0.0/16",
                    "Gateway": "172.18.0.1"
```